

# **Design Patterns in Clean Code**

Lecture notes



# In these lecture notes

We will:

- Introduce the need for software design patterns in clean code
- Introduce some design patterns



# What is clean code?

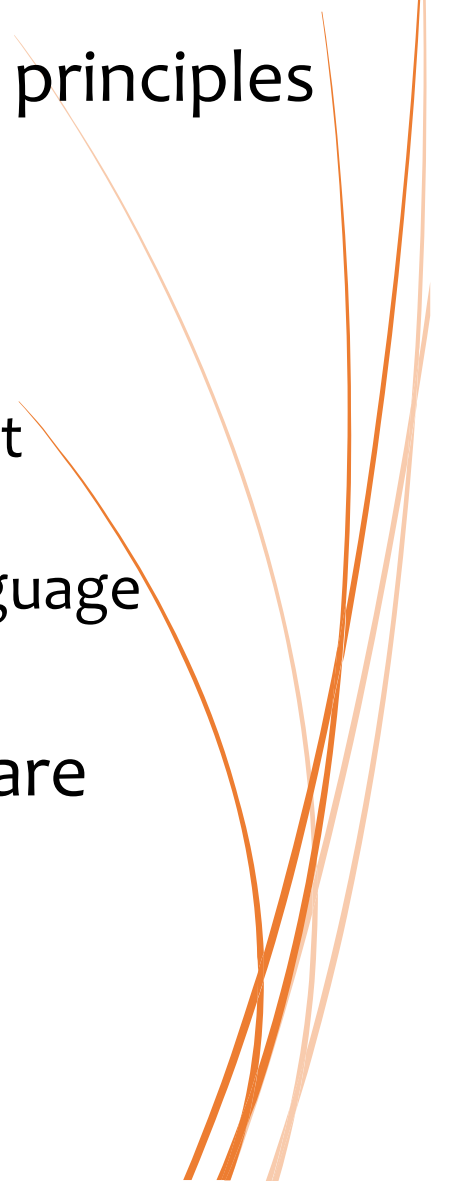
- Some of the many definitions...
    - “Clean code is code that has been taken care of. Someone has taken the time to keep it simple and orderly. They have paid appropriate attention to details. They have cared.” (Robert C Martin)
    - “Programming is the art of telling another human what one wants the computer to do. (Donald Knuth)  
“Good programmers write code that humans can understand.” (Martin Fowler)
    - “Follow standard conventions. Reduce complexity as much as possible.” (Robert C Martin)
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# Software design patterns in clean code

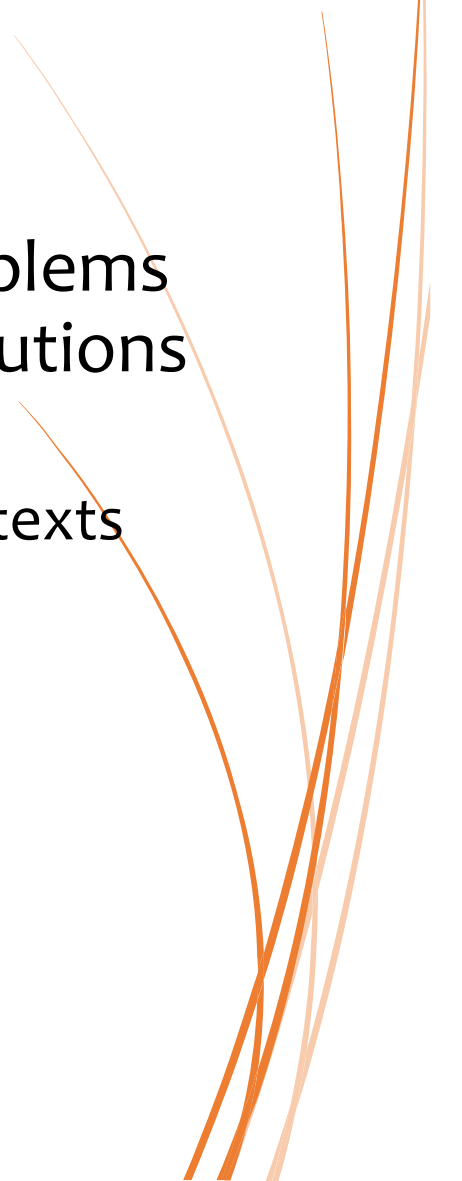
- Therefore, clean code...
  - is code that humans can understand
  - follows standard conventions
  - reduces complexity
  - is as simple as possible
  - is orderly
  - requires care
- Use of software design patterns can help to achieve all these aspirations



# Design patterns

- A design pattern captures good design principles
  - A design pattern provides a lexicon of abstractions
    - the name of a design pattern conveys a lot of meaning
    - just like the word *bike* does in natural language
  - Use of design patterns is evidence of care being taken with the code
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# Design patterns help

- Designing clean code is not easy, but...
  - Experts have recognised recurring problems and have provided tried-and-tested solutions
    - These are design patterns
    - They can be reused in many different contexts
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- Several thin, curved orange lines are positioned on the right side of the slide, extending from the top right towards the bottom right, creating a decorative, organic feel.

# Design patterns

- A design pattern is a generalised description consisting of:
  - A name for the pattern
  - The problem the pattern addresses
  - A solution to the problem
  - The consequences of using the solution
- Using a design pattern often means
  - customising the general solution...
  - to satisfy the specific requirements of...
  - a specific instance of the general problem

# Design patterns this week

- This week's lecture notes introduce the following patterns
    - Façade
    - Data Transfer Object
    - Table/View Data Gateway
    - Factory Method
    - Command
    - Null Object
  - Introduced in Server-Side Programming (SSP)
    - If you took SSP, read the notes to remind yourself of these patterns
    - Otherwise, read the notes to learn about these patterns
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